AP PRECALCULUS

Unit 1 • Lesson 1.9-1.10 • Notes (filled in)

Topic 1.9-1.10 • Rational Functions and Asymptotes

Notes (filled in)

Learning goals

- Locate vertical asymptotes and holes of a rational function from its factored form.
- Describe behavior near a vertical asymptote with one-sided limits, determining signs.
- Find the coordinates of a hole.

Vocabulary. vertical asymptote, hole, one-sided limit, $\lim_{x\to a^-}$, $\lim_{x\to a^+}$, multiplicity, simplified form.

1. Vertical asymptotes or holes?

Key idea

Factor numerator and denominator completely. For a zero a of the denominator:

- if $(x-a)$ appears __more__ times in the denominator than in the numerator, $x=a$ is a __vertical asymptote__;
- if it appears __at least as many__ times in the numerator, there is a __hole__ at $x=a$.

2. Behavior near a vertical asymptote

Near $x=a$, $|r(x)|$ grows __without bound__. To decide $+\infty$ or $-\infty$, check the __sign__ of each factor just left and just right of a. If $(x-a)$ remains an __odd__ number of times in the denominator, the sign flips across the asymptote; an __even__ number of times, it stays the same.

Example 1

Describe $r(x)=\dfrac{x+1}{(x-2)^2}$ and $s(x)=\dfrac{x-1}{(x+2)(x-3)}$ near their vertical asymptotes.

r: numerator near $3>0$, $(x-2)^2>0$ on both sides: $\lim_{x\to 2^-} r=\lim_{x\to 2^+} r=+\infty$.
s at -2: numerator near -3, $x-3$ near -5. For $x<-2$ the denominator is positive, so $s<0$: $-\infty$; for $x>-2$: $+\infty$.
s at 3: numerator near 2, $x+2$ near 5. Left: $-\infty$; right: $+\infty$.

3. Holes

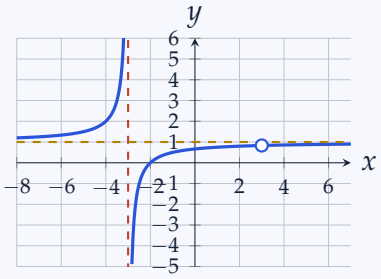
Key idea

To find the hole at $x=a$: cancel the common factors, then substitute a into the __simplified__ expression. The hole is at $(a,$ that value$)$, and $\lim_{x\to a} r(x)$ equals __that value__.

Example 2 • Full analysis

Analyze $r(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$: hole, vertical asymptote, zero, end behavior. Then sketch.

$r(x) = \dfrac{(x-3)(x+2)}{(x-3)(x+3)}$. Hole at $x = 3$: $\frac{5}{6}$, so $\left(3, \frac{5}{6}\right)$. Vertical asymptote $x = -3$: $\lim_{x \to -3^-} r = +\infty$, $\lim_{x \to -3^+} r = -\infty$. Zero $x = -2$. Horizontal asymptote $y = 1$.



Example 3 • A hole on the axis

$r(x) = \dfrac{(x-1)^2}{x-1}$. Is $x = 1$ a zero?

No. $r(x) = x - 1$ for $x \neq 1$, but $r(1)$ is undefined. The graph has a hole at $(1, 0)$, not a zero.

AP exam tip

Justify each feature with the factored form *and* a limit: "$\lim_{x \to -3^+} r(x) = -\infty$, so $x = -3$ is a vertical asymptote"; "$\lim_{x \to 3} r(x) = \frac{5}{6}$ and $r(3)$ is undefined, so there is a hole at $\left(3, \frac{5}{6}\right)$."

Common mistakes

- Calling every zero of the denominator a vertical asymptote without checking for cancellation.
- Finding the hole's height in the original expression ($\frac{0}{0}$).

Check your understanding

1. $\dfrac{x-3}{x^2-9}$ has a vertical asymptote at $x = -3$ and a hole at $\left(3, \frac{1}{6}\right)$.
2. $\lim_{x \to 0^-} \dfrac{2}{x^3} = -\infty$.